USDA Pesticide Residue Analysis

An Examination of Pesticide Detections in the U.S. Food Supply, 1994–2024

Data Source: USDA Pesticide Data Program (PDP)

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Project Overview

This project analyzes pesticide residue data from the USDA Pesticide Data Program (PDP) spanning 1994 through 2024. The PDP is one of the most comprehensive government-run residue monitoring programs in the world, annually testing thousands of produce samples for hundreds of pesticide compounds. The goal of this analysis is to inform the public about which pesticides are most commonly detected in the U.S. food supply, whether any detected compounds raise health concerns based on current scientific literature, and what the broader context of pesticide exposure looks like beyond dietary residue alone. The analysis concludes with an examination of evidence-based solutions, including organic consumption, regenerative agriculture, and microbial alternatives, as well as a discussion of what the data suggests for consumers, policymakers, and the agricultural sector.

Data and Methodology

The dataset consists of approximately 600,000 rows of detection records drawn from the USDA PDP public data files. Each row represents a single pesticide detection on a single produce sample and includes the commodity tested, pesticide name, country of origin, concentration detected, and an indicator of whether the detection exceeded the applicable EPA tolerance level. After cleaning, the dataset contained about 164,112 unique samples across 365 unique pesticide compounds, with produce samples averaging close to three pesticide detections per sample across the full dataset.

Data cleaning was performed using Python (pandas). Steps included: removing records with multiple or ambiguous countries of origin to ensure cleaner reporting; standardizing pesticide name fields where minor variations in labeling existed across collection years; and filtering to fields relevant to this analysis (commodity, pesticide name, concentration, EPA tolerance flag, and collection year). Tableau was used for dashboard development and visualization.

For the chemical research component, the ten most frequently detected pesticides in the 2019–2024 window were identified and individually reviewed against publicly available toxicology data from the NIH PubChem database, EFSA peer review conclusions, and the EPA pesticide fact sheet library. Regulatory approval status in the EU was confirmed via EFSA documentation.

The Python dataset can be located at this github repository for further review [Placeholder for when I add this]

Limitations

This analysis draws exclusively from USDA PDP data and does not incorporate data from the FDA Total Diet Study or state-level monitoring programs, which test different commodities and may show different residue profiles. The PDP's non-random sampling design (it prioritizes commodities consumed in large quantities by children) means the findings are not fully representative of all produce available in the U.S. market. Additionally, the PDP alternates which produce items it includes in testing each year, which limits the amount of information available on any specific commodity when evaluating individual years in isolation. The EPA tolerance levels for several compounds have changed over the dataset's time range, which can affect how violations are categorized across years and complicates longitudinal comparison. The compound screening list has also expanded considerably over the 30-year span of this dataset, meaning that increases in detection counts over time may partially reflect improvements in analytical sensitivity rather than actual increases in pesticide application. Lastly, the chemical research component of this project reviewed publicly available summaries of regulatory risk assessments and peer-reviewed literature. It is not a systematic meta-analysis and is not intended to be read as a comprehensive toxicological review.

Key Findings

Across the full 30-year dataset, 0.73% of samples contained detections that exceeded applicable EPA tolerance limits and it represents roughly 1,200 samples with confirmed exceedances. The more striking finding may be the residue density: the average produce sample contained an average of roughly three distinct compounds and this remained consistent across the entire study in addition to looking at years individually.

Focusing on the 2018–2024 period, the ten most frequently detected pesticides were: Fludioxonil, Thiabendazole, Azoxystrobin, Pyrimethanil, Fluopyram, Pyraclostrobin, Imidacloprid, Methoxyfenozide, Acetamiprid, and Boscalid. Of these ten compounds, only Thiabendazole, has had its food additive use banned in the EU, Australia, and New Zealand, though it remains approved as a pesticide in EU agriculture under renewed conditions. The remaining nine are currently approved in both the US and EU, though several have received scrutiny in recent years.

The Ten Most Detected Pesticides: Mechanisms and Regulatory Status

Broad-Spectrum Fungicides

Six of the ten compounds (Fludioxonil, Azoxystrobin, Pyraclostrobin, Boscalid, Fluopyram, and Pyrimethanil) are broad-spectrum fungicides that are primarily metabolized through the liver's detoxification system. In toxicological studies, these compounds share a recognizable pattern at high experimental doses. There has also been evidence of their disruption to the thyroid. Regulatory reference values for all six compounds are set well below dose levels that produce these effects in animal studies.

Of particular note is Fludioxonil, which the European Food Safety Authority (EFSA) formally classified as an endocrine disruptor in November 2024, determining that it meets the criteria for interference with estrogen, androgen, and steroidogenesis (EAS) pathways in both humans and non-target organisms.(EFSA, 2024) Earlier laboratory research had identified fludioxonil as an antiandrogen and demonstrated its ability to stimulate genes that are associated with breast cancer cell activity. (Scippo et al., 2012) This classification has significant implications for fludioxonil's continued EU approval status, which remains under review.

A concern shared across this fungicide group is the concept of metabolic competition. Because these compounds rely on overlapping enzyme pathways, simultaneous exposure to multiple fungicides may increase the strain on the liver. A 2022 NIH review on cumulative pesticide risk assessment noted that current regulatory frameworks may underestimate mixture-based risk by evaluating compounds in isolation. (Rider et al., 2022, NIH/NIEHS)

Thiabendazole: The EU Food Additive Ban

Thiabendazole occupies a distinct category within this group. While it is still approved as an agricultural fungicide in the EU, its use as a food additive (applied post-harvest to citrus skins and bananas) is banned in the EU, Australia, and New Zealand. The EPA has acknowledged in its own fact sheet that thiabendazole is likely carcinogenic at doses high enough to affect the thyroid hormone balance, while noting it is not likely carcinogenic at lower doses. The thyroid and liver are identified as the primary target organs, with animal studies showing increased organ weights in both (EPA, 2002).

In 2022, EFSA concluded following a peer review that thiabendazole formally meets the criteria for endocrine disruption specifically for the thyroid in humans. This designation elevated thiabendazole to a critical area of concern in the EU regulatory framework. Current toxicological reference values are considered protective against these effects at typical dietary exposure levels, but the review identified data gaps that prevented finalizing the assessment for non-target organisms. (EFSA, 2022)

Neonicotinoids: Imidacloprid and Acetamiprid

The two neonicotinoid insecticides in the top ten, Imidacloprid and Acetamiprid, are distinct from the fungicide group. Both activate receptors (nAChRs) in the nervous system, a mechanism that is highly selective for invertebrate receptors but not entirely without effect on mammalian neurology. A study published in PLOS ONE found that both compounds produced effects on the receptors in developing mammalian cerebellar neurons at concentrations greater than 1 micrometer leading authors to conclude that neonicotinoids may adversely affect human health, especially the developing brain. (Kimura-Kuroda et al., 2012)

A 2024 review of EPA developmental neurotoxicity (DNT) study data found that high-dose offspring of rodents exposed to imidacloprid and acetamiprid showed significant reductions in brain tissue volume, with reported effects for acetamiprid including reduced auditory startle reflex across all dose groups. The authors noted deficiencies in EPA's analytical framework and flagged that the agency had dismissed statistically significant adverse effects in several instances. (Sass et al., 2024)

There is also continued concern among researchers and specialists regarding the detrimental effects of neonicotinoids on honey bee populations. A 2025 study found that even small amounts of these pesticides can harm bees by disrupting brain function, weakening immune response, and causing physiological stress (Ahsan et al., 2025).

Methoxyfenozide

Methoxyfenozide operates through a different mechanism than the other nine compounds. As an insect growth regulator, it mimics the molting hormone ecdysone in insects, triggering premature and lethal molting. Its selectivity for insect hormonal biology means it has very limited direct interaction with mammalian biology, and it is generally considered among the lower-risk compounds in this group from a human health perspective (U.S. Environmental Protection Agency, 2024).

Pesticide Issues Extend Beyond Residue

Pesticide Mixtures

An unresolved question raised by this data is not the risk posed by any individual compound but the cumulative effect of the mixture of pesticides that consumers are routinely exposed to. Regulatory tolerances are set on a compound-by-compound basis, but food consumption involves simultaneous exposure to multiple residues at once.

Research on pesticide mixture toxicity demonstrates that interactions on metabolic processes are the most common mechanism of synergism, particularly when compounds share biological targets or detoxification pathways. A review published in Archives of Toxicology noted that cholinesterase-inhibiting insecticides, triazole fungicides, and pyrethroid insecticides are overrepresented in identified combinations, and that how the body processes a substance is a far more common source of interaction than two compounds simply amplifying each other's effects. Critically, the same review found that experimentally confirmed interactions at dietary exposure levels remain rare and have mostly occurred at unrealistically high concentrations in laboratory settings (Solecki et al., 2017).

The NIH National Toxicology Program has advocated for expanding cumulative risk assessment frameworks beyond compounds that share a single mechanism of action to include chemicals that target a common end, such as liver steatosis or endocrine disruption, even when their individual mechanisms differ.(Rider et al., 2022)

Current evidence does not demonstrate that the pesticide residues found in this dataset accumulate in the human body over time or create a predictable cumulative toxicity at typical dietary exposure levels. However, as the NIH/NIEHS review acknowledges, the science of mixture risk assessment is still developing, and the current regulatory paradigm of single-compound evaluation may not fully capture the biological reality of how these compounds interact during repeated, low-dose dietary exposure cycles.

Pesticides In The Water

Dietary exposure is only one pathway. Pesticides applied to agricultural land leach through soil profiles into groundwater and travel via surface runoff into lakes, rivers, and drinking water reservoirs. A major USGS assessment of over 1,200 public supply wells across the United States found that pesticide compounds were detected in 41% of wells, with nearly two-thirds of those containing compound mixtures rather than a single pesticide. Degradate compounds, the breakdown products of pesticides after environmental transformation, were detected more frequently than the parent compounds themselves and are often subject to fewer regulatory benchmarks. (Bexfield et al., 2021, Environmental Science & Technology)

A systematic review published found that pesticide residues in groundwater pose long-term chronic health risks and that once deep aquifers are contaminated, decontamination is exceptionally difficult. The review identified agricultural runoff, leaching, and spray drift as the primary contamination pathways, and noted that soluble pesticides are particularly mobile during precipitation events. This means that a person who eats exclusively organic food may still carry a pesticide body burden from their water supply, their local environment, or occupational proximity to conventional agriculture. (Acharya, L.K., 2025)

Pesticide Resistance

A critical dimension of the pesticide issue is pest resistance. Just as bacteria evolve resistance to antibiotics under selection pressure, agricultural pests evolve resistance to insecticides, fungicides, and herbicides when those compounds are applied repeatedly over large areas. Entire chemical mode-of-action groups have already become ineffective against certain pest species as a result of widespread resistance. The evolutionary dynamics of resistance are well understood, but management strategies remain largely reactive, implemented only after resistance has developed and spread in field populations rather than proactively before failure occurs. (Ramsden et al., 2019)

As resistance spreads and established chemical classes lose efficacy, the agricultural industry faces pressure to develop more potent, broader-spectrum compounds to maintain crop yields. This escalation cycle carries the risk of introducing compounds with less-established safety profiles into the food supply at precisely the moment when scientific understanding of mixture effects and long-term low-dose exposure is still developing. The concern is not hypothetical: it mirrors the pattern seen with the neonicotinoids, which were widely adopted as replacements for older insecticide classes and are now themselves under significant regulatory scrutiny. (Hamed et al., 2025)

Possible Solutions

The findings in this analysis do not exist in isolation. The pesticides detected in the USDA PDP data reach people not only through food but through the broader environment, including water supplies and occupational contact. Addressing the issue requires action at multiple levels, from individual consumer choices to large-scale shifts in how we grow food.

Organic Consumption

The most consistently supported individual intervention in the research literature is switching to an organic diet. A landmark study at Emory University found that when children were switched from conventional to organic diets for just five days, urinary metabolites for malathion and chlorpyrifos dropped to nondetectable levels almost immediately and remained undetectable until conventional food was reintroduced. (Lu et al., 2006).

A 2023 meta-analysis of 50 population-based studies found that organic food intake had a beneficial association with reducing pesticide exposure biomarkers and was also associated with reduced obesity risk, though evidence for specific disease outcomes was assessed as insufficient due to limited study size. (Mie, A., et al. 2023) A separate peer-reviewed study found that switching four American families to an exclusively organic diet for six days reduced detected pesticide chemicals. (Carly Hyland, et al 2019)

Regenerative Agriculture and Non-Toxic Alternatives

Individual purchasing decisions, while meaningful, cannot solve a systemic agricultural problem. The most promising large-scale alternative in current research literature is regenerative agriculture, a farming approach that prioritizes soil health, biodiversity, and reduced synthetic inputs. A 2025 NIH-published review found that regenerative organic agriculture systems produced measurable increases in vitamin C, zinc, and polyphenols in crops including leafy greens, grapes, and carrots, alongside reductions in nitrate and pesticide residues compared to conventional farming. (Seabrook, R., et al. 2025) A 2022 study in PeerJ found that regenerative farming practices enhanced the nutritional profiles of crops by restoring soil microbial communities disrupted by conventional tillage and synthetic pesticide applications. (Montgomery et al., 2022)

The fundamental principles of regenerative agriculture, keeping soil covered, minimizing disturbance, maintaining living roots year-round, increasing crop diversity, and limiting synthetic compounds, are designed to build natural pest resistance through ecosystem resilience rather than chemical suppression. A 2023 literature review in Sustainability found that while the evidence base is still growing and many growers remain cautious about adoption due to profitability uncertainty, the core mechanisms by which regenerative practices improve soil health and reduce chemical dependency are well-supported. (Khangura et al., Sustainability, 2023)

Microbial biopesticides represent another promising non-toxic avenue. Unlike broad-spectrum chemical pesticides, microbial pesticides act selectively against specific pests while minimizing harm to non-target organisms including humans. Critically, they also reduce the risk of resistance evolution by introducing biological rather than purely chemical selection pressure on pest populations (Rosier et al., 2025). With global pesticide use increasing by 50% over the past three decades, microbial alternatives offer a pathway toward breaking that escalation cycle (Hamed et al., 2025).

Conclusions

The USDA PDP data tells a story that is more nuanced than the headlines around pesticide residues typically suggest. The overwhelming majority of sampled produce, over 99%, tested within established EPA tolerance limits across the 30-year dataset. At the same time, the near-universal presence of multiple pesticide residues per sample raises legitimate scientific questions about the adequacy of single-compound risk frameworks for evaluating real-world dietary exposure.

Of the ten most commonly detected compounds, Fludioxonil received a new endocrine disruptor classification from EFSA in 2024, and Thiabendazole had already met EFSA's thyroid disruption criteria in 2022. Imidacloprid and Acetamiprid carry growing evidence of developmental neurotoxicity concerns at high doses, with findings from regulatory rodent studies challenging earlier assumptions about the safety margins of both compounds.

The research literature does offer constructive pathways forward. Peer-reviewed evidence supports organic diet transitions as an effective means of reducing measurable pesticide exposure at the individual level, while regenerative agriculture represents a scalable systemic alternative that addresses chemical dependency at the source. These solutions are not mutually exclusive and their effects are likely compounding. Public datasets like the USDA PDP exist precisely so that researchers, policymakers, and consumers can engage with the evidence rather than the headlines. What this analysis ultimately reflects is a need for more rigorous cumulative risk science, more transparent regulatory communication, and sustained investment in agricultural practices that prioritize long-term safety alongside productivity.

References

Acharya, L. K., Paramaguru, P. K., Tripathi, K., et al. (2025). Pesticide contamination in groundwater: Processes, risks, and mitigation strategies. *Discovery Agriculture, 3*, 152. https://doi.org/10.1007/s44279-025-00337-x

Ahsan, Z., Wu, Z., Lin, Z., Ji, T., & Wang, K. (2025). The Sublethal Effects of Neonicotinoids on Honeybees. Biology, 14(8), 1076. https://doi.org/10.3390/biology14081076

Bexfield, L. M., et al. (2021). Pesticides and pesticide degradates in groundwater used for public supply across the United States. *Environmental Science & Technology*. https://pubs.acs.org/doi/10.1021/acs.est.0c05793

European Food Safety Authority. (2013). *Scientific opinion on the developmental neurotoxicity potential of acetamiprid and imidacloprid*. https://www.researchgate.net/publication/307678062

European Food Safety Authority. (2022). Peer review of the pesticide risk assessment for the active substance thiabendazole in light of confirmatory data submitted. *EFSA Journal, 20*(4), e7212. https://pmc.ncbi.nlm.nih.gov/articles/PMC8958497/

European Food Safety Authority. (2024). Peer review of the pesticide risk assessment of the active substance fludioxonil. *EFSA Journal, 22*(11), e9047. https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2024.9047

Hamed, A. S. A., & Hidayah, N. (2025). Global trade and pesticide use. In B. R. Babaniyi & E. E. Babaniyi (Eds.), *The interplay of pesticides and climate change*. Springer. https://doi.org/10.1007/978-3-031-81669-7_5

Hyland, C., Bradman, A., Gerona, R., et al. (2019). Organic diet intervention significantly reduces urinary pesticide levels in U.S. children and adults. *Environmental Research, 171*, 568–575. https://doi.org/10.1016/j.envres.2019.01.024

Khangura, R., et al. (2023). Regenerative agriculture: A literature review on the practices and mechanisms used to improve soil health. *Sustainability, 15*(3), 2338. https://www.mdpi.com/2071-1050/15/3/2338

Kimura-Kuroda, J., Komuta, Y., Kuroda, Y., Hayashi, M., & Kawano, H. (2012). Nicotine-like effects of the neonicotinoid insecticides acetamiprid and imidacloprid on cerebellar neurons from neonatal rats. *PLOS ONE, 7*(2), e32432. https://pubmed.ncbi.nlm.nih.gov/22393406/

Lu, C., et al. (2006). Organic diets significantly lower children's dietary exposure to organophosphorus pesticides. *Environmental Health Perspectives*. https://pmc.ncbi.nlm.nih.gov/articles/PMC1367841/

Mie, A., et al. (2023). The effects of organic food on human health: A systematic review and meta-analysis. *PubMed*. https://pubmed.ncbi.nlm.nih.gov/37930102/

Montgomery, D. R., et al. (2022). Soil health and nutrient density: Preliminary comparison of regenerative and conventional farming. *PeerJ, 10*, e12848. https://peerj.com/articles/12848/

Ramsden, M. W., et al. (2019). The evolutionary origins of pesticide resistance. *Biological Reviews*. https://pmc.ncbi.nlm.nih.gov/articles/PMC6378405/

Rider, C. V., McIntyre, B. S., & Foster, P. M. D. (2022). Mixture math: Deciding what to add in a cumulative risk assessment. *Environmental Health Perspectives, 130*(9). https://pmc.ncbi.nlm.nih.gov/articles/PMC9262140/

Rosier, C. L., et al. (2025). From soil to health: Advancing regenerative agriculture for improved food quality and nutrition security. *Frontiers in Nutrition*. https://pmc.ncbi.nlm.nih.gov/articles/PMC12576041/

Sass, J. B., Donley, N., & Freese, W. (2024). Neonicotinoid pesticides: Evidence of developmental neurotoxicity from regulatory rodent studies. *Frontiers in Toxicology, 6*, 1438890. https://doi.org/10.3389/ftox.2024.1438890

Scippo, M. L., et al. (2012). Endocrine disruptors fludioxonil and fenhexamid stimulate miR-21 expression in breast cancer cells. https://www.researchgate.net/publication/232228772

Seabrook, R., et al. (2025). Regenerative organic agriculture and human health: The interconnection between soil, food quality, and nutrition. https://pmc.ncbi.nlm.nih.gov/articles/PMC12108233/

Solecki, R., Kortenkamp, A., Bergman, A., et al. (2017). Toxicological interactions of pesticide mixtures: An update. *Archives of Toxicology, 91*, 3211–3237. https://pubmed.ncbi.nlm.nih.gov/28845507/

U.S. Environmental Protection Agency. (2002). *Pesticide fact sheet: Thiabendazole*. https://www3.epa.gov/pesticides/chem_search/reg_actions/reregistration/fs_PC-060101_1-May-02.pdf

U.S. Environmental Protection Agency. (2024). *Methoxyfenozide 2SC* [Pesticide product label, EPA Registration No. 83100-65]. https://www3.epa.gov/pesticides/chem_search/ppls/083100-00065-20240131.pdf